Physics 30 Worksheet # 9: *Charged Particles in Electric Fields*

1. An electron is accelerated from rest across a potential difference of 85.0 V. What is the final speed of the electron?

2. An electron is accelerated from rest to a speed of 6.00×10^{6} m/s. What potential difference is required to do this?

3. An electron is accelerated through a potential difference of 9.00 V. If the electron begins at a speed of 6.00×10^{4} m/s, what is the final speed of the electron?

4. A proton is accelerated from a speed of 5.00×10^{4} m/s to 5.00×10^{6} m/s. What potential difference is required to accomplish this?